INDIVIDUAL PROJECT REPORT

Review 1 — Problem Identification & Proposal (24/8/2026)

AI-Driven Edge–Cloud Network for Cyclone Prediction and Resilient Ocean Telemetry

Student	Register No.	Team No.	My role
Sreehari	24BDS0417	_____	Sensing layer (sensor and satellite simulators, hardware nodes) and recovery of lost data

Team: Laksh Ahuja (24BDS0212), Anvay Sharma (24BCE0824), Sreehari (24BDS0417). **Repository:** https://github.com/laksh-ahuja06/AI-DRIVEN-EDGE-CLOUD-NETWORK

Summary

A cyclone-monitoring network is only as good as the data its sensors deliver. In this report I describe the problem from the sensing side: realistic sensor readings, the lossy links they travel over, and how lost readings can be recovered. I present the evidence, the 15 articles I studied, the proposed solution and my plan up to Review 2.

1. Problem Statement

Ocean buoys and coastal stations lose packets in bursts when storms bring rain fade and interference, so the monitoring centre receives gaps in the very readings that show a cyclone intensifying.

- **Why it matters:** the pressure trend is a key sign of intensification; a gap hides it.
- **Who is affected:** forecasters who interpret the data, and coastal communities whose warnings depend on it.
- **If unsolved:** dashboards show missing or stale values during the storm, and models that need continuous data cannot run.

Objectives of My Part

Objective	Measurable success criterion
Realistic sensor readings with a storm that arrives and passes	Wind > 60 km/h and pressure < 990 hPa only during the storm
Packet loss that matches real marine links	Configurable loss rate; bursts of consecutive losses
Record the true value of every reading, even lost ones	Ground-truth file for every test run
Recover lost readings better than simple methods	Lower error than holding the last value
Low-cost hardware node sending the same message format	≤ ₹1,000 per node

2. Evidence the Problem is Real

Table 1: Evidence that the problem is real

Source	Published fact	Why it matters for my part
India Meteorological Department [1]	India has a coastline of about 7,500 km; the 1999 Odisha super cyclone caused about 10,000 deaths.	Coastal sensing covers a long, exposed coastline.
WMO Atlas 1970–2019 [2]	Tropical cyclones are among the leading causes of human and economic losses from weather, climate and water extremes.	Data gaps during storms have real human cost.
Engineering estimate (our calculation)	Low-power long-range radio (LoRa) offers only about 0.3–50 kbps [5]; marine links also suffer correlated (bursty) loss [8].	Sensor data must be compact and the receiver must cope with bursts of lost packets.

3. Literature Survey

Table 2: Literature survey (15 articles)

#	Author / Year	Method used	Result	Limitation
1	Al-Fuqaha et al., 2015 [3]	Survey of IoT technologies	Protocols and architectures for sensors	General, not marine
2	Xu et al., 2019 [4]	Review of marine IoT monitoring	Ocean sensing platforms and links	Data loss handling not addressed
3	Raza et al., 2017 [5]	LPWAN overview	Range vs data rate trade-offs	No recovery of lost data
4	Augustin et al., 2016 [6]	LoRa measurements	Long range at low throughput	Duty-cycle limits
5	Jouhari et al., 2019 [7]	Underwater sensor network survey	Challenges of marine communication	Underwater focus
6	Gilbert, 1960 [8]	Two-state burst-noise	Models clustered errors	Theory only

#	Author / Year	Method used	Result	Limitation
		channel		
7	Elliott, 1963 [9]	Error rates on burst channels	Practical channel parameters	Not for IoT telemetry
8	Che et al., 2018 [10]	GRU-D with missing-value decay	Better results with gaps	Offline clinical data
9	Cao et al., 2018 [11]	Bidirectional RNN imputation	State-of-the-art gap filling	Needs whole series; heavy
10	Yoon et al., 2018 [12]	GAN-based imputation (GAIN)	Imputes without full data	Complex to train
11	Lim & Zohren, 2021 [13]	Deep time-series forecasting survey	Forecasting model families	No real-time telemetry
12	Holland, 1980 [14]	Hurricane pressure profile	Realistic storm pressure field	Needs storm parameters
13	Chen et al., 2019 [15]	CNN on satellite channels	Intensity from imagery	Large data and compute
14	Denby & Lucia, 2020 [16]	Orbital edge computing	Process data where captured	Space segment only
15	Shelby et al., 2014 [17]	CoAP for constrained devices	Compact sensor messaging	No gap recovery

Gap: imputation methods (GRU-D, BRITS, GAIN) work offline on complete datasets with heavy frameworks; marine IoT reviews describe links but not recovery; burst-loss channel models are rarely used to test sensor systems realistically. My part will generate realistic readings and loss, record the truth so recovery can be measured, recover lost readings online, and design low-cost nodes that send the same data.

4. Proposed Solution Form

Form: coding and modelling first, low-cost fabrication later. Simulated sensor nodes and a simulated satellite station give us unlimited, repeatable storm scenarios with known ground truth, which is essential for measuring recovery. Once the software works, ESP32 nodes with a pressure sensor will replace the simulator using the same message format. A product design was rejected because the scope is a proof of concept, not a manufactured device.

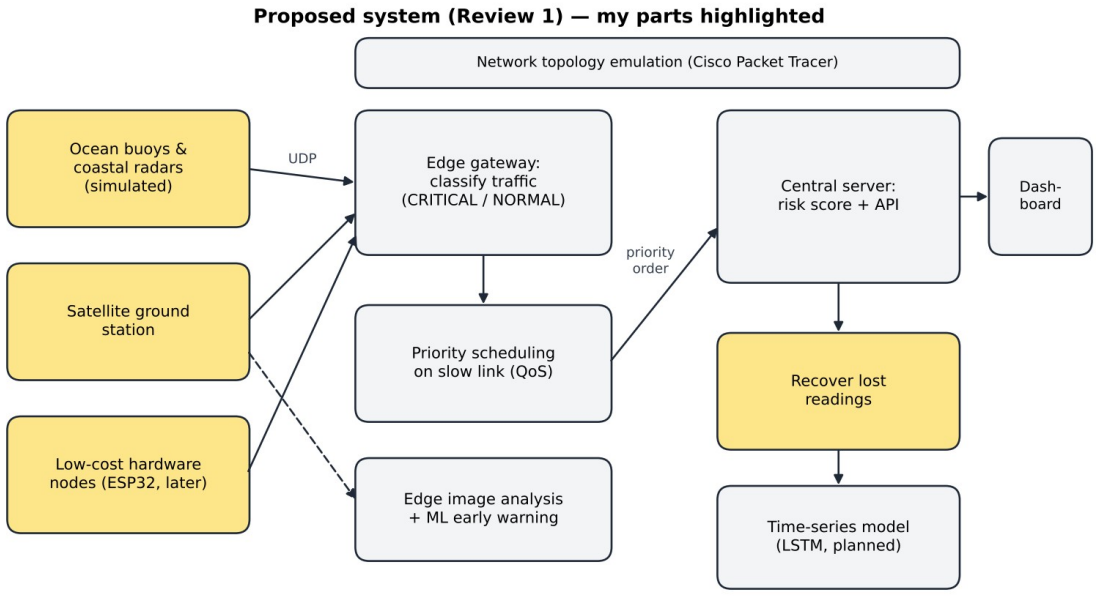


Figure 1: Proposed block diagram; my parts (sensors, satellite station, hardware nodes, recovery of lost readings) are highlighted

Planned Method for My Part

Component	Planned method
Sensor simulator	One reading per second with a sequence number, compact JSON over UDP, and a configurable loss probability.
Storm and loss models	Begin with a ramp, then a physical wind/pressure profile; begin with random loss, then a burst-loss channel.
Recovery of lost readings	Linear interpolation as the baseline, then a learned time-series model evaluated against recorded ground truth.
Hardware	ESP32 with a BMP280 pressure/temperature sensor sending the same JSON as the simulator.

5. Initial Technical Feasibility

Table 3: Feasibility of my part

Area	Status
Skills I have	Python, basic electronics, Arduino programming, statistics
Skills I must learn	Burst-loss channel modelling, time-series reconstruction, ESP32 Wi-Fi/UDP programming, I²C sensors
Tools needed / access	Python (free); Arduino IDE (free); ESP32 + BMP280 sensor (about ₹500 per node, college lab may lend); laptop
Risks	No real buoy data (mitigation: physics-inspired simulation); hardware parts delayed (mitigation: software first); recovery may not beat simple interpolation

Table 4: Risk register for my part

Risk	Likelihood	Impact	Mitigation
Hardware parts delayed or faulty	Medium	Medium	Software simulator first; borrow from lab
Simulated data too unrealistic	Medium	High	Use published storm profiles and burst-loss models
Recovery no better than interpolation	Medium	Medium	Report honestly; keep interpolation as fallback
Pressure reads wrong indoors (altitude)	High	Medium	Convert station pressure to sea level

Initial cost estimate for hardware (R1): about ₹2,000–2,500 for two nodes including breadboards and cables; software costs nothing.

6. Team Role Split and Timeline to Review 2

Table 5: Team role split

Member	Role	Specific tasks (R1 → R3)
Laksh Ahuja (24BDS0212)	Central server, dashboard & network QoS	Prediction server and REST API, live dashboard, priority scheduling on the slow link, cyclone simulation model, demo launcher, automated testing
Anvay Sharma (24BCE0824)	Edge gateway & edge intelligence	Gateway classification and forwarding, satellite image analysis at the edge, ML early-warning classifier, LSTM implementation, evaluation and logs
Sreehari (24BDS0417)	Sensing layer, hardware & data recovery	Sensor-node and satellite-station simulators, packet-loss model, reconstruction of lost readings, model training data, ESP32 hardware nodes

My role is the beginning of the data path and its repair: what the sensors send, how it gets lost, and how we recover it.

Table 6: My weekly plan from Review 1 to Review 2

Week	Dates	My milestone	Deliverable / check
1	25–31 Aug	Sensor-node simulator: readings, sequence numbers, configurable loss	Node sends JSON over UDP; LOST lines printed
2	1–7 Sep	Cyclone scenario (wind rises, pressure falls) and gap-filling baseline on the server	Missing sequence numbers are filled and marked
3	8–14 Sep	Integration test with gateway and server; hardware parts list	Prototype shown at Review 2; component list ready

R1 Video Plan

Segment	What I show
Problem and evidence (1.5 min)	Face + screen; IMD / WMO evidence and the problem statement
Literature gap (1.5 min)	Face + screen; literature table and the gap
Proposed solution and feasibility (2 min)	Face + screen; block diagram and feasibility table
My role and timeline (1 min)	Face + screen; role table and weekly plan

References

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